

CS 2490 – Homework 1

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You **must** turn in your assignment as a PDF document entitled: *last-name_firstname_hw1.pdf*. You can use L^AT_EX or any other formatting tool you want, or you can write your solutions by hand and scan them. But the resulting document must be a PDF file.

Exercise 1. Use induction to prove the following statement:

$$\forall n \in \mathbb{N}, \sum_{i=0}^n i^2 = \frac{n(n+1)(2n+1)}{6}.$$

Exercise 2. Consider the following statement: “In any non-empty box of crayons, all crayons have the same color.”

We propose the following proof of the statement, reasoning by induction¹ on the number n of crayons in the box.

Proof. Base case: $n=1$. For a box with only 1 crayon, then surely all the crayons in that box have the same color.

Induction step. Suppose the property is true for boxes containing n crayons. Consider a box with $n+1$ crayons. Select the first n crayons: by the induction hypothesis, they all have the same color, say c . Select the n last crayons: by the induction hypothesis, they all have the same color, say c' . Since the crayons in the middle have color c and c' at the same time, it forces $c = c'$ and so all crayons have the same color in the end. $\square$

¹ The statement is in plain English, but notice that it has the correct form for induction to be used on it, since we want to prove something “for all number $n \geq 1$ of crayons”.

If you have ever seen a box of crayons before in your life, then you know that the statement cannot possibly be true. Spot and explain the mistake in the proof by induction above.

Exercise 3. Let Σ be an alphabet, and w be a word over it. Recall that the word w^n is the word w concatenated with itself n times:

$$w^n := \underbrace{w \cdot w \cdots w}_{n \text{ times}}.$$

In other words, $w^0 = \varepsilon$ and for all $n \in \mathbb{N}$, $w^{n+1} = w^n w$.

Using induction, prove the following statement:

$$\forall n \in \mathbb{N}, |w^n| = n|w|.$$

Recall that $|u|$ denotes the length of a word u , i.e., the number of letters in u .

Exercise 4 (*). In this exercise, we work over the alphabet $B = \{0, 1\}$. Let L be a language over B where each word has the same number of 0s and 1s. Prove that any word in L^* also has the same number of 0s and 1s.

Hint: Identify this statement as being of the form $\forall n \in \mathbb{N}, P(n)$ for a property P that you will make explicit, and then use induction.

If you have trouble reasoning with an abstract language L , you can first assume that L is the following language: $\{1001, 010101, 100101\}$. You should then realize that your proof doesn't involve the specific definition of this L , but rather the properties it satisfies. Then carry on in translating the proof to an abstract L .